20 kW / 80 kWh Hybrid Solar Power System

20.46 kWp PV array | 20 kW three-phase hybrid inverter | 81.92 kWh LiFePO4 storage

Quote & Client Details

Quote No.	TPV-261026-001	Date	26-Oct-2026	Valid Until	25-Nov-2026
Client Name		Contact No.		Client Email	
Site Address		Supply Type	Three-phase 415 V	VAT TIN No.	_____

20.46 kWp PV array capacity	**20 kW** Inverter output (3-phase)	**81.9 kWh** Battery storage	**USD 5,000** Total system cost
86 kWh Avg. daily generation	**31.5 MWh** Est. annual generation	**74 kWh** Usable storage (90% DoD)	**17 t** CO2 avoided per year

1. Executive Summary

Tororo PV Power Company Limited is pleased to submit this proposal for the design, supply, installation and commissioning of a **20 kW / 80 kWh hybrid solar power system** for the client's residence. The system combines a **20.46 kWp** rooftop array of 33 Jinko 620 W N-type bifacial modules, a Deye 20 kW three-phase hybrid inverter and 16 Dyness 5.12 kWh LiFePO4 battery modules, giving the house clean daytime power and up to **74 kWh** of usable night-time and back-up energy.

The array is expected to generate about **31,541 kWh per year** (approx. 86 kWh/day). At an indicative grid tariff of UGX 800/kWh this is worth about **UGX 22.7 million per year** in avoided electricity costs. The turnkey price is **USD 5,000** (UGX 19,400,000), based on USD 250 per kW of system capacity, including equipment, mounting, balance of system, logistics, installation and commissioning.

Key Benefits

- Uninterrupted power: automatic switch-over to solar + battery in under 10 ms during grid outages.
- Lower bills: most daytime load and much of the evening load supplied from the sun.
- Long life: Tier-1 N-type modules (30-year output warranty) and 6,000+ cycle LiFePO4 batteries.
- Remote monitoring: inverter Wi-Fi logger with mobile app for live generation, load and battery data.
- Scalable: the inverter and battery bank can be expanded or paralleled later.

2. Equipment Schedule (Bill of Materials)

#	Item	Specification	Qty	UoM
1	Hybrid inverter	Deye SUN-20K-SG05LP3-EU, 20 kW, 3-phase 400 V, 48 V battery, 2 MPPT, Wi-Fi logger	1	pcs
2	Battery modules	Dyness DL5.0C, 5.12 kWh, 51.2 V 100 Ah LiFePO4 (total 81.92 kWh)	16	pcs
3	Battery racks	19-inch cabinet racks, 8 modules each, with busbars & comms cables	2	pcs
4	Solar PV modules	Jinko JKM620N-66HL4M-BDV, 620 W N-type bifacial (total 20.46 kWp)	33	pcs
5	Mounting structure	Anodised aluminium rails, mid/end clamps, roof hooks / L-feet	1	system
6	DC protection	PV combiner with DC isolators, 1000 V fuses & Type II SPD; battery DC breaker/fuse	1	set
7	AC protection	AC distribution board, MCBs, RCD, Type II AC SPD, change-over switch	1	set
8	Cabling	6 mm² PV solar cable, 70 mm² battery cable, AC cable, MC4 connectors, conduit & trunking	1	lot
9	Earthing	Copper earth rods, earth cable and bonding of array frame and enclosures	1	set

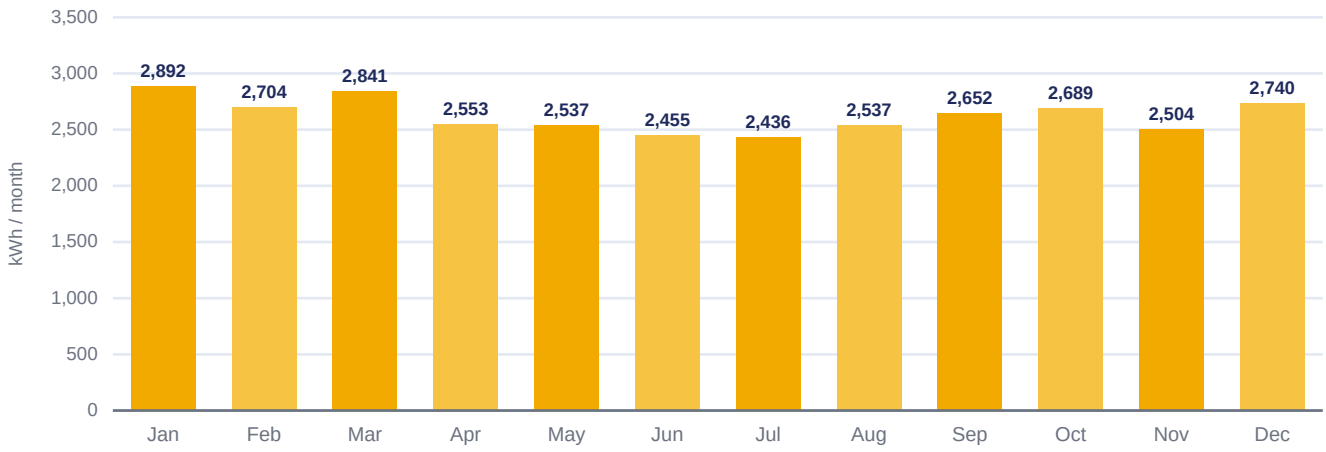
3. PV Array & String Design

Parameter	Value
Modules	33 x 620 W = 20.46 kWp
String configuration	3 strings x 11 modules
MPPT 1 / MPPT 2	2 strings / 1 string
String Voc (STC)	approx. 551 V
String Vmp (STC)	approx. 462 V
DC/AC ratio	1.02
Roof area used	approx. 95 m²

Battery parameter	Value
Modules	16 x 5.12 kWh (all in parallel)
Nominal voltage	51.2 V DC
Total / usable energy	81.92 / 73.7 kWh
Chemistry	LiFePO4, BMS with CAN comms
Design cycle life	6,000+ cycles @ 90% DoD
Back-up @ 3 kW load	approx. 25 hours
Back-up @ 5 kW load	approx. 15 hours

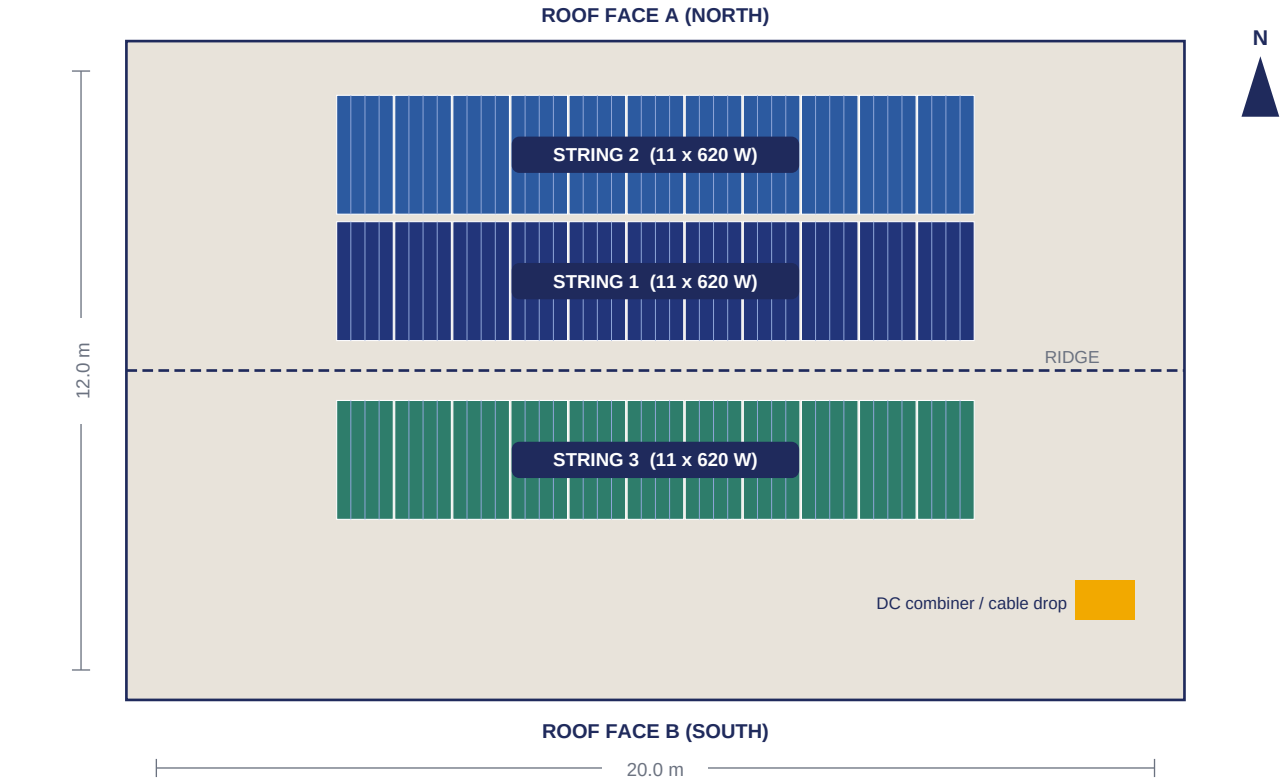
4. Estimated Energy Generation

Estimate based on long-term average irradiation for Tororo (lat. 0.69° N) and a performance ratio of 80%. Annual total: **31,541 kWh**.

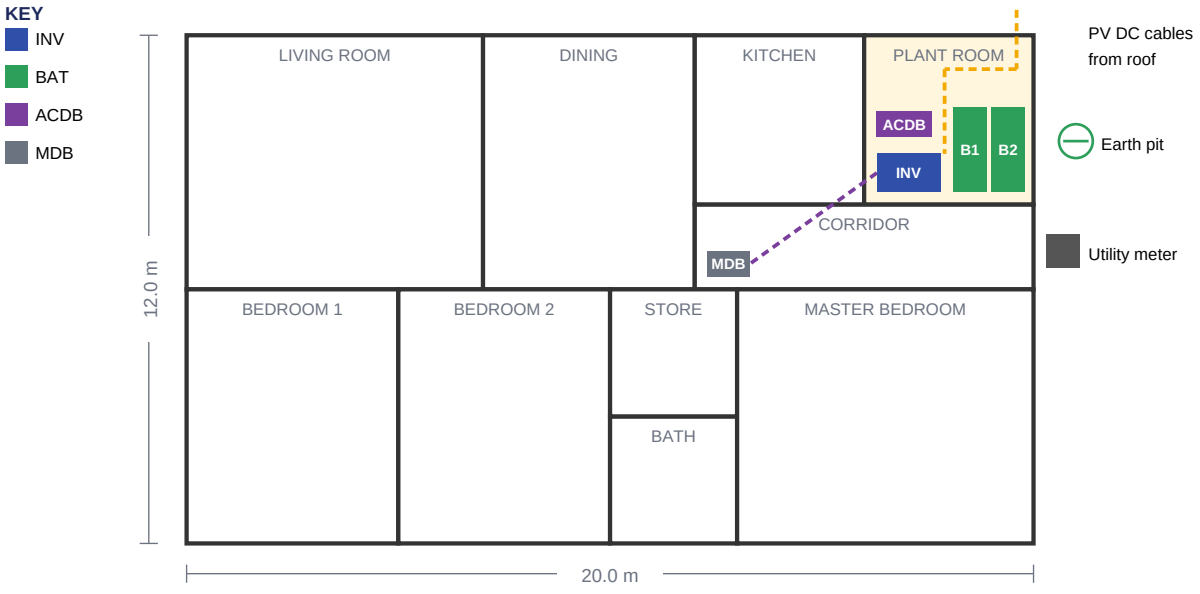


5. Roof Plan - PV Array Layout

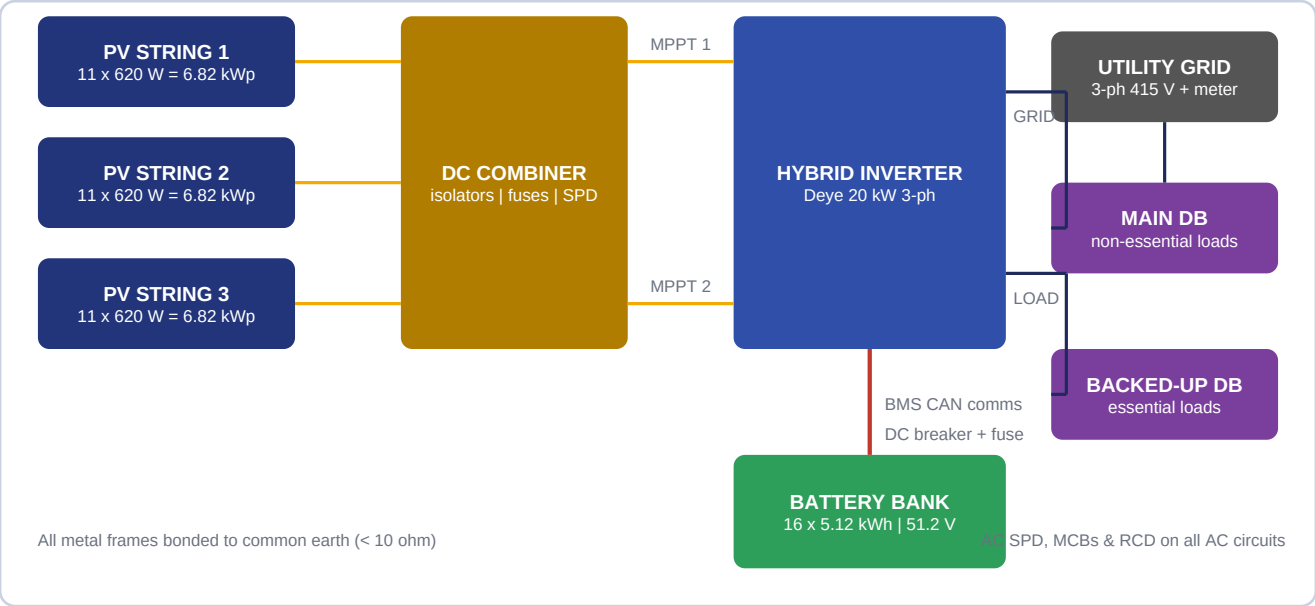
Indicative layout on a typical 20 m x 12 m pitched-roof bungalow. Final layout will be confirmed after the site survey (roof orientation, shading and structure).



6. Ground Floor Plan - Equipment Location & Cable Routes



7. Single-Line Diagram



8. Implementation Schedule

Phase	Activity	Duration
1. Site survey	Roof inspection, load audit, shading & supply check, design confirmation	1-2 days
2. Procurement	Equipment ordering, delivery to site, pre-installation checks	1-2 weeks
3. Installation	Mounting structure, modules, inverter, batteries, cabling, earthing & protection	4-5 days
4. Commissioning	Insulation & polarity tests, settings, monitoring app set-up, handover	1 day
5. Training	Client training on operation, app monitoring and basic care	2 hours

9. Standards, Warranty & After-Sales Support

Component	Warranty
PV modules	12 yrs product / 30 yrs power
Hybrid inverter	5 years (manufacturer)
LiFePO4 batteries	10 years (manufacturer)
Installation workmanship	2 years

Installation follows IEC 62548 (PV array design), IEC 60364-7-712 and the Uganda Electricity Regulatory Authority wiring requirements. After handover we provide remote monitoring support, a free 6-month check-up visit and optional annual maintenance (module cleaning, torque checks, firmware updates, battery health report).

10. Commercial Offer

Item	Qty	UoM	Cost (UGX)
Deye 20 kW 3-phase hybrid inverter (SUN-20K-SG05LP3-EU)	1	pcs	Included
Dyness 5.12 kWh 51.2 V 100 Ah LiFePO4 (DL5.0C)	16	pcs	Included
Jinko 620 W N-Type bifacial (JKM620N-66HL4M-BDV)	33	pcs	Included
Mounting structure	1	system	Included
Balance of system (protection, cabling, earthing, racks)	1	system	Included
Total Material Cost			**16,697,600**
Logistics & transport			Included
Commercials			Included
Installation labour, testing & commissioning			Included
Total Additional Costs			**2,702,400**
Subtotal (excl. VAT)			19,400,000
Less discount (not applied)			0
VAT (not applicable)			0
TOTAL SYSTEM COST (UGX)			**19,400,000**
TOTAL SYSTEM COST (USD)			**5,000.00**

Pricing basis: 20 kW x USD 250/kW = USD 5,000; exchange rate UGX 3,880/USD. Prices valid until 25-Nov-2026.

11. Financial Summary (indicative)

Indicator	Value	Indicator	Value
Annual generation	31,541 kWh	Annual savings	UGX 22,709,618
Self-consumed share	90%	Simple payback	approx. 10 months
Tariff assumed	UGX 800/kWh	25-year savings (no escalation)	UGX 568 M

12. Payment Details & Terms

Payment Method	Account Number	Bank / Branch	Account Name
Bank Transfer	_____	_____	Tororo PV Power Company Limited
Mobile Money	_____	_____	Tororo PV Power Company Limited

- Deposit to commence: 100% of total material cost + 50% of additional costs (commercials, labour & logistics); remaining 50% of additional costs due on completion.
- Installation warranty: 2 years on workmanship; manufacturer warranties apply to equipment.
- Site inspection required before final quotation confirmation; a three-phase supply is assumed.
- Additional costs may apply for non-standard installations, roof reinforcement or site conditions.

13. Acceptance & Authorization

By signing below, the client acknowledges acceptance of this proposal and the terms outlined above.

CLIENT

Name: _____

Signature: _____

Date: _____

FOR TORORO PV POWER COMPANY LIMITED

Name: _____

Signature: _____

Date: 26-Oct-2026 _____

Thank you for considering Tororo PV Power Company Limited for your solar energy needs.